

Introduction

The brain's instantaneous perception of the natural world is a long-understood phenomenon, yet its neurological basis remains a scientific mystery to the field of visual neuroscience. Centuries of neuropsychology research, from lesion studies in non-human primates (Hubel, D. H., & Wiesel, T. N., 1959) to modern neuroimaging techniques (Zaretskaya, 2021), have revealed a remarkable degree of specialization in the visual cortex. As such, researchers have cemented the idea of cortical modularity, where distinct brain regions are dedicated to processing different aspects of visual information (Haynes, 2006).

Studies of specialized modules in the visual cortex have motivated researchers to investigate the functional organization of the brain. For instance, past research has established the fusiform face area (FFA) and parahippocampal place area (PPA) as brain regions specialized for face perception and scene perception, respectively (Kanwisher, 1997; Epstein & Kanwisher, 1998). Discoveries of separate modules and their unique functions provide pivotal insights into how visual information flows and is processed throughout the cerebral cortex.

Our increasing knowledge of the structure of the visual cortex gives rise to a fundamental question: How does the brain become remarkably selective in processing visual information? Understanding the mechanisms of visual processing may support existing theories of how cortical modules evolve in adaptation to the environment. For example, studying the functional connectivity of face-selective regions may explain human adaptations to social complexity (Sato et al., 2016). Similarly, investigating connectivity in scene-selective regions may reveal neural adaptations to the visual and navigational demands of the surrounding environment (Epstein et al., 2007). Further research into the specialized modules and pathways within the visual cortex is necessary to answer such questions about the evolutionary and developmental origins of brain function. In this paper, we will provide our current understanding of visual scene perception and its neural bases, and discuss opportunities to delve deeper into the special interactions that facilitate this process.

Categorizing Visual Representations

Many neuroimaging studies of visual categorization have particularly distinguished object and scene perception (Troiani et al., 2014). When viewing a particular environment, such as a kitchen, the brain takes in a variety of visual elements that could be categorized into one or the other. While focal attention may be drawn to a central object, such as a human chef, peripheral vision can capture the presence of surrounding furniture, such as a microwave, as well as the walls enclosing the space.

The simultaneous perception of both objects and scenes activate various brain regions that take in different levels of visual information. Low-level features are basic visual attributes extracted directly from sensory input. For objects, these include colors, shapes, and textures; for scenes, these include spatial layout, relative distances between elements, and variations in lighting. By contrast, high-level features refer to more complex and abstract attributes that are successively extracted through higher-level processing. These may involve the identity, contextual association, and semantic information related to the object or scene. Low-level and high-level features are processed in a hierarchical manner, giving rise to separate cortical pathways that each play a distinct, but equally significant role in visual perception (Groen et al., 2017).

Although they are used frequently in neuroscience research, it is important to acknowledge the ambiguity of the terms “object” and “scene” perception. Visual features shared across both objects and scenes make it difficult to fully distinguish the two as exclusive categories. For instance, a chair is both an object and a component of the scene’s spatial layout, creating considerable overlap between patterns of activation. Additionally, the separation between scene and object-selective regions is not strictly modular in terms of their anatomical locus (Haxby et al., 2001). Moving forward, understanding how these intertwined processes influence visual perception will require a closer examination of the representational structure of their respective pathways.

Representational Structure of the Ventral and Dorsal Pathways

We now examine the process in which both object and scene information is relayed through the visual processing stream. At first glance, low-level visual information from the retina travels to the primary visual cortex (area V1) in the occipital lobe, where neurons decode motion, color, and spatial (orientation) information (Braddick 2001). The computation of these simple visual aspects give rise to two distinct, but highly interactive pathways: the ventral (“what”) pathway at the bottom, and the dorsal (“where”) pathway at the top (Ungerleider & Mishki, 1982; Goodale & Milner, 1992; Haxby et al., 1994; Milner, 2017). Although both originate from a common source, each relay different information for a separate, unique function.

The ventral pathway, extending through retinotopically-organized visual areas (V2, V4) to the inferior temporal cortex (ITC), primarily carries object-specific information to aid in object recognition and identification. Lying inside of the ventral pathway are several regions specialized in visual identification. Neuroimaging studies reveal that the lateral occipital complex (LOC) activates strongly towards any intact three-dimensional shape, which suggests a crucial role in intermediate-level object perception (Malach et al., 1995; Kim, 2009; Margalit, 2016). From there, anterior regions in the fusiform gyrus, such as the fusiform face area (FFA) and visual word form area (VWFA), take on higher-level visual processing and, more specifically, object

recognition (Kanwisher et al., 2022). The ventral pathway essentially works at a hierarchical level; as visual input travels through a series of cortical regions, successively higher-level visual features are extracted and pieced together to identify the presented object.

Above the ventral pathway lies the dorsal pathway, which leads into the posterior parietal cortex (PPC). Regions along the dorsal pathway are known to carry information about navigation and spatial location, guiding movements that rely on visual perception of space (Goodale and Milner, 1992). Several scene-selective regions have been associated with the dorsal pathway, though their anatomical locations are ambiguous. The occipital place area (OPA), for example, responds to scenes at the level of local, scene-related elements (i.e. walls, floors, ceilings) regardless of their spatial boundary or arrangement (Kamps, 2016). The retrosplenial cortex (RSC), lying in the parietal medial temporal pathway (Kravitz et al., 2011) but holding strong connectivity to dorsal, retrieves spatial information and facilitates mental navigation (Epstein, Parker, et al. 2007). Overall, the dorsal pathway primarily processes spatial information and guides behavioral decisions in response to the surrounding space.

Representational Structure of Scene-Selective Regions

As explored above, object and scene information diverge to different locations and processes. Compared to its counterpart, however, scene processing stands out due to its broader functionality across multiple pathways. The three primary scene-selective regions are OPA, RSC, and PPA. While the dorsal pathway hosts OPA and RSC, both vital for spatial perception, the third scene-selective region, PPA, lies in between the ventral and dorsal pathways; specifically, it is situated in the medial temporal lobe.

The roles of these regions in scene perception has been a matter of much debate, particularly with PPA. One study found that scenes with a greater number of contextual associations were shown to elicit significantly higher responses from PPA within the parahippocampal cortex (PHC), suggesting that contextual associations are the dominant modulation for PPA activation in both indoor and outdoor settings (Bar et al., 2008). Later findings, however, have proposed alternative explanations for PPA activation, suggesting responsiveness to the basic awareness of 3D space, which exists without the presence of contextual associations (Mullally & Maguire, 2011). Although PPA can be categorized as a “scene-selective” region, its distinct contribution to scene perception remains unclear.

Nonetheless, several studies have made surprising observations about the capabilities of these scene-selective regions, some of which exhibit shared characteristics among themselves and with other brain regions. In a fMRI study, Harel et al. (2013) tested the presence of furniture objects against both nonspatial and spatial backgrounds. Both PPA and LOC elicited greater responses to

scenes with objects than without, indicating a shared ability to decode object presence and identity. On the contrary, RSC showed no sensitivity to the presence or absence of objects, but did share increased responses to the presence of spatial layout with PPA. Uniquely, PPA was the only region that decoded nonspatial backgrounds, unlike the general insensitivity to nonspatial factors found in both LOC and RSC. In another study by Kamps et al. (2016), researchers found that OPA shows no sensitivity to changes in spatial boundary, such as empty versus fractured rooms, but instead responds to local elements within the scene, such as surfaces and furniture. This is in stark contrast to PPA and RSC, which both reflect the global spatial arrangement of a scene, such as an empty but arranged room, without regard to the individual elements of the spatial boundary.

As demonstrated by these properties, both spatial and object information are not entirely separate in visual processing. Despite structural differences, some specialized modules share preferences for certain visual features with other regions. Such findings provide compelling evidence for interaction between the ventral and dorsal pathways.

Bi-directional Interactions between the Ventral and Dorsal Pathways

Based on recent neuroimaging studies, interactions between scene and object processing are bidirectional and suggest a mutually beneficial relationship between the ventral and dorsal pathways. One explanation for this interaction is contextual integration. Investigating the role of objects in facilitating scene processing, Brandman & Peelen (2023) used magnetoencephalography (MEG) to test representations of three main stimuli categories: blurry scenes and objects, blurry scenes alone, and objects alone. Their data showed that the presence of an object sharpened perception of blurry scenes in the left scene-selective visual cortex (left OPA and left PPA) from 300 ms after stimulus onset, suggesting that objects facilitate scene processing after the initial scene categorization takes place. It is possible that the ventral and dorsal pathways interact at a categorical stage, sharpening perception in both directions. To investigate this topic further, Weismann et al. (2023) presented scenes with single objects, finding that single objects are often sufficient in aiding scene categorization. However, they acknowledge that the contextual effectiveness of an object varies based on its diagnosticity; objects that are uniquely associated with the particular scene are deemed more useful for scene categorization. Interestingly, the presence of an incongruous or less contextually-associated object actually interferes with scene representations (Joubert et al., 2007). This research proves that objects influence scene processing at a contextual level.

Evidence also suggests that scenes facilitate the processing of objects, verifying their bidirectional relationship. In fact, some propose that scenes have a greater impact on object processing than vice versa, due to their more simple and stabilized nature (Brandman & Peelen,

2023). Putting this hypothesis to the test, Smith & Federmeier (2024) manipulated scene-preview durations when measuring participants' responses to scene-object pairs using electroencephalogram (EEG). The study found that longer scene-preview durations correlated to more accurate object categorization, suggesting that the presence of a scene facilitates object processing at a delay. Additionally, they conclude that the facilitation becomes more rapid as the object-scene pair becomes more "naturally co-occurring", or contextually associated. With recent findings, it is clear that scene-selective and object-selective regions are highly parallel and interactive in support of visual perception.

Anatomical Connectivity between the Ventral and Dorsal Pathways

Anatomical connectivity, or the structural connections throughout the cortex, provides strong evidence for the interaction between object-selective and scene-selective regions. Previous studies have found that the pathway connected to a respective region is a strong determinant of its representational abilities. Particularly, the dorsal pathway contributes to spatial background and scene representation, while the ventral pathway is significantly involved in object representation. Regions connected to both pathways, like PPA, carry both object and scene information to some extent (Harel et al. 2013).

Studies observed that RSC, which has strong connectivity to the dorsal pathway but weak connectivity to the ventral, responds to the presence of spatial layouts but not objects. On the contrary, LOC, solely connected to the ventral pathway, responds to objects and shapes, but is significantly weakened by the presence of spatial information. PPA, connected to both the dorsal and ventral pathway, is sensitive to the presence or absence of both objects and spatial backgrounds, and also carries a unique ability to decode the absence of space (Harel et al., 2013). Additionally, direct cortical streams along the ventral and dorsal pathways observed strong white matter connectivity, aiding distinct yet rapid interactivity between object and scene processing (Huang et al., 2021). As demonstrated, although the ventral and dorsal pathways serve different functions, they share and exchange specific aspects of visual information to complete their respective tasks. Future research should investigate the finer details and dynamic changes in anatomical connectivity, which may help bridge the neural gap between visual perception and higher-level cognitive functions.

Functional Connectivity between the Ventral and Dorsal Pathways

As explained, the patterns of anatomical connectivity among scene- and object-selective brain regions provide some explanations for the functional characteristics of those regions. However, investigating the extent to which those regions are functionally connected during visual perception requires more attention.

Previous studies that have measured the functional connectivity between scene-selective regions have found that the posterior network (OPA and posterior PPA) synchronizes with the early visual regions during low-level visual processing, and the anterior network (RSC and anterior PPA) synchronizes with the higher-level scene regions (i.e. the hippocampus) during more complex functions like navigation and memory (Baldassano et al., 2016). A recent, comprehensive study has verified these patterns, while also finding preferential communication between the ventral and dorsal pathways (Watson & Andrews, 2024).

While this new body of research has outlined the broad functional and structural connectivity among scene-selective brain regions, studies have yet to address the dynamic shifts that inevitably occur during naturalistic viewing. Watson & Andrews (2024) analyzed connectivity within scene networks using both the Game of Thrones and StudyForrest datasets, comparing rest periods to movie-watching tasks. Thus, they have found a specific pattern of functional connectivity that can be generalized across the entire movie. A future study using a similar experimental design could focus on comparing connectivity patterns across different scenes within the movie, which would uncover finer trends between both object and scene-selective regions. Particularly, we hypothesize that the functional connectivity between PPA and the ventral and dorsal pathways is dynamic, and is modulated by the changing spatial dimension presented in the stimuli. A way to test this hypothesis would be to compare the viewing of indoor scenes and outdoor scenes. One prediction is that PPA shows stronger synchronization with RSC during outdoor scene perception, given their shared responsiveness to spatial aspects of scenes like the expanse dimension (indoor, outdoor) (Kravitz et al., 2011; Kamps et al., 2016). On the contrary, PPA is predicted to show stronger synchronization with both OPA and LOC during indoor scene perception, as these regions respond more strongly to the presence of objects within scenes (Harel et al., 2013; Kamps et al., 2016).

Studying the functional connectivity between object and scene-selective regions could reveal valuable information about how information flows through the visual stream. A previous finding provides evidence of a feedback loop during visual perception, in which information from the dorsal pathway is relayed to the ventral pathway to aid in object processing (Brandman and Peelen, 2017). This verifies the interactions between distinct object and scene related pathways, specifically through feedback processes. Yet, current findings still do not address the coordination of activity within these networks, nor provide insight into dynamic or temporal shifts. Researchers should aim to answer existing questions about these functional interactions in future experiments.

Conclusion

Visual perception is an instantaneous and complicated process, one that has not been fully uncovered. As visual inputs travel to the primary visual cortex, they split into two distinct neural streams: the ventral pathway and dorsal pathway. Researchers have explored the different roles of object and scene-selective regions along each pathway, in order to better understand how visual information flows through the cortex. These studies provide valuable knowledge of the unique functions that each region serves, including LOC, PPA, OPA, and RSC. Discussions of their roles help complete the visual puzzle and further the neural understanding of “perception.”

As the field continues to establish and define separable cortical modules in the visual cortex, it becomes increasingly important to study the interactions between them. Previous studies prove that object and scene processing have a bidirectional relationship, in which both categories facilitate perception of the other. These studies give rise to the idea of anatomical connectivity, which many researchers have proven exists throughout both ventral and dorsal pathways. In comparison, the study of functional connectivity in the visual cortex has only recently emerged, emphasizing the need for further research into region-specific interactions using naturalistic stimuli.

Delving into visual perception presents a significant opportunity to unravel one of neuroscience's greatest mysteries. We must seek to hone and develop our understanding of the neurological underpinnings that give rise to our daily perception of the world around us. Doing so will eventually paint a vivid portrayal of visual perception at a neural basis.

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